

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – C Code Output Prediction

|                  |                                                                                                               |               |                                              |
|------------------|---------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------|
| Course Code:     | CSA0305                                                                                                       | Course Title: | Data Structures                              |
| CO Assessed:     | CO1 – Imitation (BL3, BL4)                                                                                    | Assessment:   | Assessment Tool 1 – C Code Output Prediction |
| Weightage:       | 35% of CIA                                                                                                    | Total Marks:  | 100                                          |
| CO1 Statement:   | Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. |               |                                              |
| Student Name:    | T.Raju                                                                                                        | Reg. No.:     | 192465038                                    |
| Section / Batch: | CSE-CS                                                                                                        | Date:         | 10/07/2026                                   |

### Code of Conduct

I T.raju (192465038)

(Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Raju.T

### Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20\*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

| Section / Topic                                                                   | Focus                                      | Q Nos  | Marks     |
|-----------------------------------------------------------------------------------|--------------------------------------------|--------|-----------|
| Linked data structure                                                             | Basic data structure                       | 1 - 2  | 20        |
| Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure | Classifications of Linear data structure   | 3 - 4  | 20        |
| Search Data Structure                                                             | Linear Search and Binary Search            | 5 -6   | 20        |
| Recursion, Performance analysis                                                   | Recursion                                  | 7 - 8  | 20        |
| Array Data Structures                                                             | Implementations Using Array data structure | 9 - 10 | 20        |
| GRAND TOTAL:                                                                      |                                            |        | 100 Marks |

### Q1 C Code – Array Traversal

```
#include <stdio.h>

int main() {
    int arr[] = {5, 10, 15, 20};
    int i, sum = 0;
    for(i = 0; i < 4; i++)
        sum += arr[i];
    printf("%d", sum);
    return 0;
}
```

### Your Answer

50

## Q2 C Code – Linear Search

```
#include <stdio.h>

int main() {
    int arr[] = {2,4,6,8,10};
    int key = 8;
    int i;
    for(i=0;i<5;i++){
        if(arr[i]==key){
            printf("%d",i);
            break;
        }
    }
}
```

### Your Answer

3

**Q3 C Code – Binary Search Mid**

```
#include<stdio.h>
int main(){
int low=0,high=7;
int mid;
mid=(low+high)/2;
printf("%d",mid);
}
```

**Your Answer**

3

**Q4 C Code – Pointer with Array**

```
#include<stdio.h>
int main(){
int arr[]={10,20,30};
int *p=arr;
printf("%d",*(p+2));
}
```

**Your Answer**

30

#### Q5 C Code – Array Update

```
#include<stdio.h>
int main(){
int arr[]={1,2,3};
arr[1]=arr[0]+arr[2];
printf("%d",arr[1]);
}
```

#### Your Answer

4

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**Q6 C Code - Linked List Node**

```
#include<stdio.h>
struct Node{
int data;
struct Node *next;
};
int main(){
struct Node n1,n2;
n1.data=10;
n2.data=20;
n1.next=&n2;
n2.next=NULL;
printf("%d",n1.next->data);
}
```

**Your Answer**

20

**Q7 C Code - Time Complexity Loop**

```
for(i=0;i<n;i++)  
printf("*");  
The complexity is
```

**Your Answer**

$O(n)$

**Q8 C Code - Nested Loop**

```
for(i=0;i<n;i++)  
for(j=0;j<n;j++)  
printf("*");
```

**Your Answer**

if  $n=3$   
\*\*\*\*\*

**Q9 C Code - Binary Search Complexity**

```
while(low<=high){  
    mid=(low+high)/2;  
}
```

**Your Answer**

$O(\log n)$

**Q10 C Code - Pointer Increment**

```
#include<stdio.h>  
int main(){  
    int a[]={5,10,15};  
    int *p=a;  
    p++;  
    printf("%d",*p);  
}
```

**Your Answer**

10